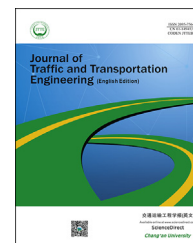


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Original Research Paper

Analysis of motorcycle microscopic characteristics at roundabouts under mixed traffic condition: A case study of Vietnam

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HIGHLIGHTS

- Two mixed traffic constitutions are found: a performance rule and the presence of small vehicles.
- Motorcycles continuously alter their speed while traveling, at a temporal resolution of 0.5 s.
- The maximum turning angle rate is estimated according to the speed.
- The critical gap of motorcycles (1.25 s) is shorter than that of lane-based vehicles.
- The motorcycle's following space has an oval shape.

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ABSTRACT

Recently, the demand for comprehension of mixed traffic in developing countries, particularly at roundabouts, which are highly interactive road junctions, has increased. Thus, we analyzed mixed traffic at roundabouts, considering Vietnam as a case study. The two main objectives of this study were to characterize the mixed traffic in Vietnam and to determine the microscopic characteristics of motorcycles at roundabouts. First, efforts were made to clarify the two constitutions of mixed traffic (the performance rule and the presence of small-sized vehicles), and the term “motorcycle-oriented mixed traffic” was defined. Even when satisfying the two fundamental constitutions, this traffic state has unique features, e.g., only one type of non-lane-based vehicle (the motorcycle) and the predominance of motorcycles in the traffic composition (91.7%). Second, four microscopic characteristics of motorcycles were obtained from a large dataset: the motorcycles’ continuous changes in speed, the relationship between the turning angle rate and the speed, the critical gap, and the following space. The relationship between the turning angle rate and the speed was first formulated as a power curve. The critical gap of motorcycles was estimated as a small value (1.25 s) in the case study. The following spaces varied with respect to the speed and had an oval shape. The smallest lateral and longitudinal dimensions were 1.5 and 1.74 m, respectively. While all the findings are meaningful, they are restricted to the case study (Ho Chi Minh City, Vietnam).

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1. Introduction

“Mixed traffic” is a popular technical term used to describe the state of traffic in developing countries (Asaithambi et al., 2016; Das and Maurya, 2018; Kiran and Verma, 2016; Lee et al., 2009; Matsuhashi et al., 2005; Vu and Shimizu, 2007; Wong et al., 2016). Mixed traffic is a complex system, even for mid-blocks or intersections. Furthermore, mixed traffic at roundabouts represents an immensely intricate situation; owing to the numerous interactions among vehicles travelling in different directions, having a wide range of characteristics, and rescinding lane discipline. They represent a non-signalized intersection that employs the yield on entry and circulation rules TRB (2010). This intricacy motivated us to investigate the characteristics of mixed traffic flow at roundabouts. Vietnam was selected as a case study, as it exemplifies the foregoing complex characteristics.

The macroscopic characteristics of mixed traffic at roundabouts were analyzed only once previously, by Lo (2017). However, the microscopic characteristics, which are essential for understanding and correctly reproducing driver behavior in simulations, are yet to be studied. In the present study, we bridged this gap by investigating the microscopic characteristics of vehicles from an individual viewpoint. There were two main objectives: 1) to characterize the uniqueness of mixed traffic in Vietnam (case study), and 2) to unveil the microscopic characteristics of motorcycles at roundabouts.

The remainder of this paper is organized as follows. Sections *Homogeneous traffic and mixed traffic* and *Constitution of mixed traffic* present the definitions and constitutions of mixed traffic. Section *Comparison between HCM city and other mixed traffic cities* describes the traffic state in Vietnam (categorized as motorcycle-oriented mixed traffic) and compares the traffic scenario with those of other developing countries. Section *Data collection method* describes the study site and outlines macroscopic characteristics of the traffic flow. Section *Results and discussion* focuses on the characteristics at the micro-level, including the acceleration, deceleration, turning angle, turning angle rate, critical gap, and following space. This section emphasizes the three key findings of the paper, which are the continuous maneuvering and changing speeds of motorcycles, the relationship between the turning angle rate and speed, and the following space of motorcycles. Finally, Section *Conclusions* presents conclusions and suggestions for future research. A flowchart of the study is shown in Fig. 1.

1.1. Homogeneous traffic and mixed traffic

Homogeneous traffic is a popular term used to describe the traffic state in developed countries. This definition includes

two fundamental characteristics: cars are the dominant vehicle type and drivers follow lane discipline. Arasan and Krishnamurthy (2008) defined homogeneous traffic state as “traffic movement under fairly homogeneous traffic conditions with cars constituting about 80% or more of the vehicles displays lane discipline.” When all vehicles obey the lane regulations, there is synchronized movement on the road.

However, the traffic state in developing countries is more complex. The increase in the types of vehicles and the changes in performance rules have led to chaos. Vehicles vary with regard to dimensions (e.g., lateral and longitudinal size) and performance capabilities (e.g., acceleration, deceleration, desired speed, and maneuvering capabilities). Many types of vehicles that are smaller than cars can gain more freedom in their lateral movement by disregarding the lane rule. By utilizing the advantages of static and dynamic characteristics, small vehicles can occupy empty space on the road surface to maximize their utility. This practice results in the behaviors reported by Lee (2007), such as filtering, swerving or weaving, tailgating, oblique following, and maintaining a shorter headway when aligning with the lateral edge of the preceding vehicle. Similarly, overtaking behavior was described by Asaithambi and Shravani (2017), and the accumulation at the stop line was described by Haque et al. (2008). This traffic stream finally results in unsynchronized movement.

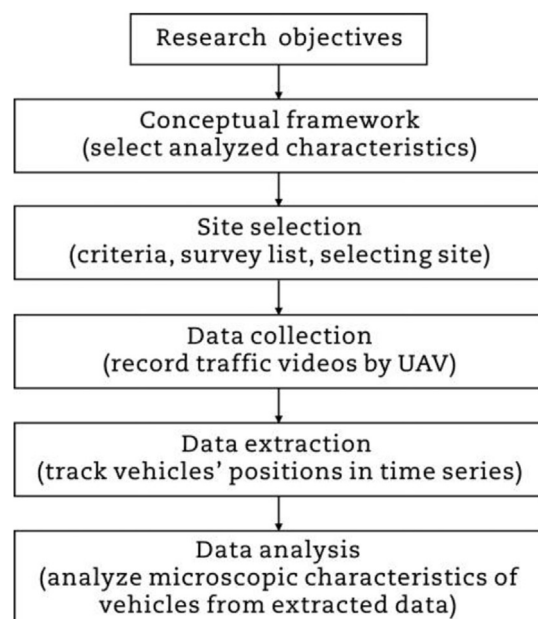


Fig. 1 – Flowchart of characteristics analysis.

1.2. Constitution of mixed traffic

Researchers have attempted to define the mixed traffic state in previous studies. Taylor and Mahmassani (1999) conceptualized heterogeneous traffic according to its mixed nature, as bicycles, motorcycles, and cars share road space. Hsu et al. (2003) described a measurement method for mixed traffic characteristics in Taiwan, which involves calculating the ratio of the number of motorcycles to the volume of traffic. It emphasizes the mix essence of motorcycles and other vehicles. Chandra and Kumar (2003) examined the mixed traffic in India, which includes various vehicles. Minh et al. (2007) studied traffic in Hanoi and Ho Chi Minh (HCM) City in Vietnam, where motorcycles account for more than 80% of transportation. It is emphasized that the mixed traffic should be considered with a sufficient motorcycle volume. Tang et al. (2009) investigated the heterogeneity of mixed traffic in China, including the vehicle heterogeneity and speed heterogeneity. In most of the aforementioned studies, mixed traffic was considered as traffic with a wide variety of vehicles having different characteristics, all sharing the road surface. Lane discipline was not considered as a critical factor affecting mixed traffic behavior in these studies.

More recently, mixed traffic has been defined as a lack of lane discipline and a variation in vehicular characteristics. Venkatesan et al. (2015) stated that large variations in speed and vehicle dimensions lead to difficulties in following lane discipline. Small vehicles often occupy any available space on the road to maximize their utility. Asaithambi et al. (2012) characterized mixed traffic in developing countries as traffic having a lack of lane discipline, varying constituent vehicle types, and a significant disparity in the characteristics of the vehicles within each class. Shiomi et al. (2012) characterized the characteristics of traffic flow dominated by motorcycles free from lane restrictions and examined the differences in the field of view, size, weight, maneuvering methods, turning radius, and acceleration and deceleration between passenger cars and motorcycles. Finally, Kiran and Verma (2016) used the terms “heterogeneous disordered traffic” and “mixed traffic” to describe the traffic scenario in developing economies. The two characteristics distinguishing this type of traffic from homogeneous traffic were identified as a wide variety of static and dynamic characteristics and the absence of lane discipline. These two factors were confirmed via microsimulation, which indicated that they have significant effects on the capacity and jam density Lenorzer et al. (2015). The combination of these two effects results in unsynchronized movement of vehicles. Poorly performing vehicles present considerable resistance to the movement of high-performing vehicles, and the entire stream becomes difficult to predict. Thus, many distinctive behaviors have been identified during surveys of mixed traffic: overtaking, which was mentioned by Asaithambi and Shravani (2017), oblique following Lee (2007), swerving movement Nguyen (2012), vehicle filtering, creeping, cut-tail, giving way, accumulation and dispersion at intersections, and many others, as identified by Trinh et al. (2018).

The foregoing literature review raised some arguments that must be clarified. First, what would the traffic state be if there was a wide variety of vehicle types, ranging from small to large, and all of them followed lane discipline? In this scenario, the type of vehicle factor is activated, and the performance rule factor is deactivated. This scenario may describe the traffic state of developed countries, such as the United States of America or Japan, where small vehicles still obey lane-based movement. Thus, it is classified as homogeneous traffic according to the traffic's synchronized movement. Second, what scenario would develop if we limited the categories of vehicles to one (four-wheelers), but rescinded the lane discipline? Here, the first factor is deactivated, and the second factor is activated. This scenario is represented by traffic at non-lane-marked roundabouts in developed countries. The homogeneous traffic state remains, and no mixed traffic characteristics are observed. This could be because four-wheelers (and larger vehicles) cannot utilize small spaces on road surfaces, owing to their limited maneuverability. The third scenario concerns limiting the categories of vehicles to one (motorcycles) and rescinding lane discipline. This scenario is represented by Vietnam, where motorcycles account for up to 80% of traffic. Most of the characteristics of mixed traffic have been observed there.

Thus, it can be concluded that the two fundamental factors causing mixed traffic are the performance rule and the appearance of small vehicles. If only one of these factors is active, traffic will not exhibit mixed traffic characteristics. Large vehicles (cars, trucks, and buses) cannot utilize small spaces on road surfaces, owing to their limited maneuverability, even when lane discipline is revoked. Additionally, small vehicles, such as motorcycles and three-wheelers, cannot perform complex maneuvers if they are forced to follow lane discipline.

1.3. Comparison between HCM city and other mixed traffic cities

To compare Vietnam with other mixed traffic cities, data on the traffic composition in mixed traffic cities were collected, as shown in Table 1. We considered the following issue: when choosing data for a country, a city, a town, or even the same road, the traffic proportions of different locations are highly diverse. For comparison, we selected the highest proportion of non-lane-based vehicles among the study sites from each country.

In accordance with the proposals of Hsu et al. (2003) and Tang et al. (2009), this paper suggests that two indicators should be used—the number of non-lane-based vehicle types and the proportion of non-lane-based vehicles in the traffic—to quantitatively characterize mixed levels in traffic. Regarding the appearance of non-lane-based vehicles, it is clear that as the number of non-lane-based vehicles increases, the traffic becomes more complex. Owing to the larger variations in their dimensions and performance capabilities, the interactions between different types of vehicles and their behaviors become more sophisticated. Regarding the proportion of non-lane-based vehicles, the

Table 1 – Comparison between traffic composition in mixed traffic cities.

City	Traffic composition (%)				Type of non-lane-based vehicle	Reference
	Motorcycle	Three-wheeler	Bicycle	Lane-based vehicle		
HCM City	89.5	–	–	10.5	Motorcycle	Trinh et al. (2018)
Ha Noi	92.7	–	–	7.3	Motorcycle	Nguyen and Sano (2012)
Taipei	35.7	–	–	64.3	Motorcycle	Wong and Liao (2012)
Tainan	36.4	–	–	63.6	Motorcycle	Lee and Wong (2016)
Penang	66.0	–	–	34.0	Motorcycle	Ahmed et al. (2015)
Pakem	87.2	0.1	1.4	11.3	Motorcycle, three-wheelers, bicycle	Hidayati et al. (2012)
Bangkok	55.3	2.6	–	43.1	Motorcycle, three-wheelers	Transport Statistics Group (2018)
Chennai	56.0	13.0	–	31.0	Motorcycle, three-wheelers	Bharadwaj et al. (2017)
Dhaka	50.3	6.0	–	43.7	Motorcycle, three-wheelers	Bangladesh Bureau of Statistics (BBS) (2019)
Kathmandu	65.3	1.7	–	33.0	Motorcycle, three-wheelers	Timalsena et al. (2017)

higher proportions, the more the characteristics of mixed traffic are revealed. According to the data in Table 1, the cities defined in this study were characterized using two indicators, as shown in Fig. 2. The horizontal and vertical axes are assigned to these two indicators.

Considering the horizontal axis of Fig. 1, there are two extremes: no non-lane-based vehicles and all non-lane-based vehicles. The former extreme is the state of synchronized traffic in developed countries. The latter extreme describes the chaotic state of traffic that has not yet been discovered. In Fig. 2, HCM City is the second-closest city to the latter extreme case; motorcycles comprise 89.5% of the traffic flow. Thus, HCM City is considered to be the representative of the latter extreme. Moreover, the city has only one type of non-lane-based vehicle (motorcycles), making it relatively easy to study motorcycle's interactions. The other cities were located between the two extremes. Hence, by understanding the mixed traffic in HCM City, the characteristics of the other mixed traffic cities can be inferred.

As shown in Fig. 2, both HCM City and Ha Noi belong to the mixed traffic group; both are characterized by the predominance of motorcycles, which characterizes the properties of the entire traffic stream. This state is close to the second aforementioned extreme, i.e., the chaotic traffic state. Priority is usually ignored owing to the high density of motorcycles. Instead, the common features are a high level of interaction and the negotiation of priority. Thus, the term

“mixed traffic” alone does not precisely describe the traffic status of Vietnam.

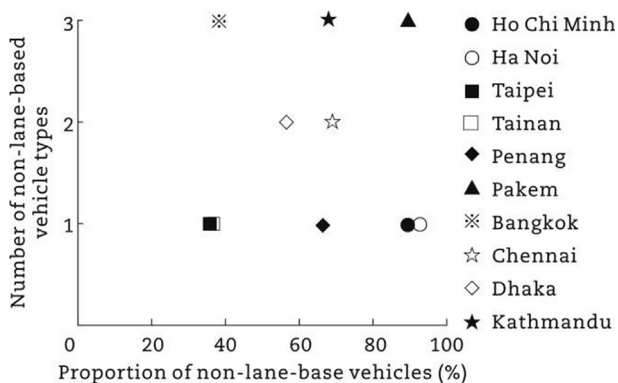
In previous studies, the predominance of motorcycles was emphasized by terming this traffic as “motorcycle homogeneous” traffic by Minh (2007), “motorcycle-dominated mixed traffic” by Minh et al. (2010), “mixed traffic with high share of motorcycle” by Huynh (2016), “non-lane based mixed traffic” by Shiomi et al. (2012), and “motorcycle-oriented mixed traffic” by Matsuhashi et al. (2005). Herein, the terminology employed by Matsuhashi et al. (2005) is used to describe the traffic context of Vietnam, for the following reasons. First, the term “motorcycle-oriented mixed traffic” highlights the presence motorcycles, which are the only non-lane-based vehicles and the primary small vehicles. Moreover, it emphasizes the adjustment of the whole mixed traffic characteristics on the basis of the features of motorcycles and the traffic flow likely to be a motorcycle stream.

2. Data collection method

2.1. Survey site

The traffic in HCM City exhibits both the aforementioned features of mixed traffic: the revocation of lane discipline and the appearance of non-lane-based vehicles. Although the city's traffic is dominated by motorcycles (>90% in this study), it is a good representative of mixed traffic. Additionally, HCM City is a major city, contributing more than half of Vietnam's gross domestic product, and is the representative city in Vietnam for a mixed traffic status. Thus, it was selected as a case study to examine the microscopic characteristics of mixed traffic.

The survey site was the small signalized roundabout in HCM City, Vietnam. The roundabout is circular in shape, with six legs for both entry and exit, as shown in Fig. 3. The diameters of the central island and inscribed circle are 15.2 and 51.7 m, respectively. It has no lane markers, which gives drivers freedom to maneuver inside the roundabout. The national driving regulations stipulate that traffic should drive in a counterclockwise circular direction and should obey the “give-way” rule, which requires drivers to yield to

**Fig. 2 – Comparison of mixed traffic cities.**

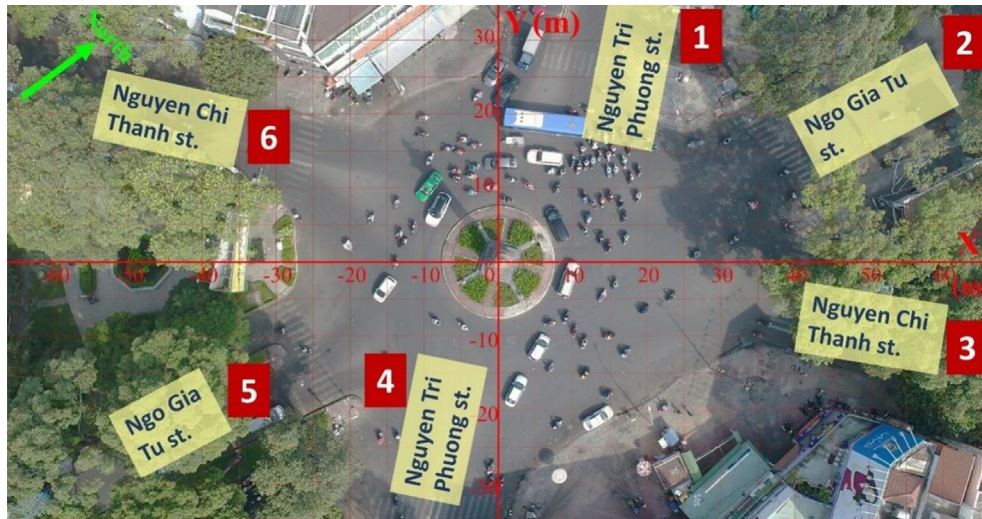


Fig. 3 – Screenshot from a recorded video.

traffic that is already in the roundabout. The study area also conforms to the right-hand traffic rule.

This roundabout was selected owing to several factors. First, its design is typical for roundabouts in HCM City, with regard to the number of legs. In developing countries, owing to the limited space in urban areas and the limited infrastructure investment for multilevel intersection options, roundabouts are frequently used where intersections with more than five legs are needed. This design is also suggested to improve the operation efficiency of the same-pave-area multi-leg intersection [AASHTO \(2011\)](#). Moreover, it is one of the main connectors of road networks. Three four-lane arterial roads, which have four-lane widths and two-way traffic, are connected by the circular roundabout. These legs likely have

similar widths, numbers of lanes, crossing angles, and traffic flow. Finally, the roundabout has the advantage of permitting a large circulating road width and can be considered as a three-lane roundabout. This provides abundant space for motorcycles to take full advantage of their mobility; hence, the behavior desired in the study is likely to occur.

Videos of traffic at the roundabout were recorded during the peak rush hour (9:00 AM to 10:30 AM) on March 1, 2018. They were recorded using an unmanned aerial vehicle (UAV): the quadcopter DJI Phantom 4 Pro. The UAV's field of view covered the whole roundabout area (132.84 m × 70.05 m), as shown in [Fig. 3](#). The dimensions of the roundabout are shown in [Fig. 4](#). The UAV's height above ground level was maintained

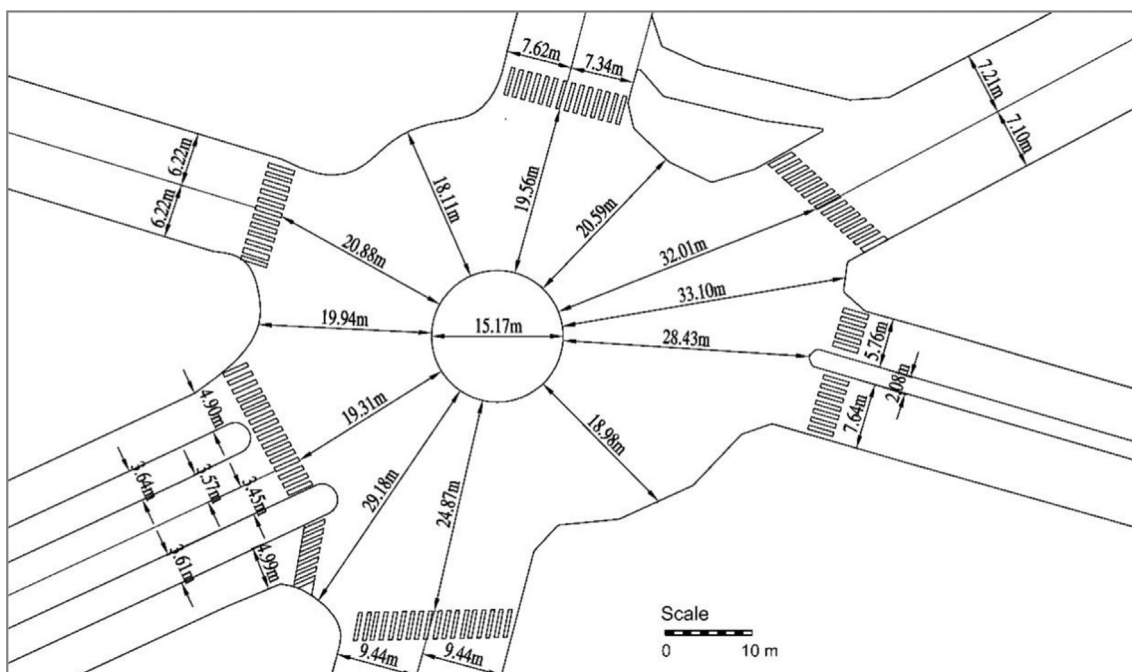


Fig. 4 – Dimensions of the survey site.

at 87 m during the recording, and the function of auto-adjust location was used to keep the UAV stationary above the center of the roundabout. To ensure sufficient details, the videos were recorded in 4 K ultrahigh-definition resolution, 4096×2160 pixels, at 60 frames per second. The spatial resolution of the captured field was approximately 0.032 m/pixel.

2.2. Data extraction

Follow a manual video-based data extraction method, a 40-inch monitor (3840×2160 pixels) was used to detect the positions of vehicles at each frame of the videos. For each period of 30 frames, the experimenter clicks once at the vehicle on the video. Based on the attached coordination system, the XY coordinates of the position are calculated. In order to eliminate the error caused by UAV's small motion, the coordination system is updated manually in accordance with the change in the scene. The final combination of all the detected locations on the XY coordinate system formed the trajectory of the vehicle's movement. Under the assumption that the human detection error on the monitor was 0.001 m, the accuracy of the extracted data was approximately 0.14 m. This value is smaller than other studies automatic data extraction method, e.g., 0.5–0.6 m (Kim et al., 2019) and 1.045 m (Apeltauer et al., 2015).

The vehicle trajectories were then extracted from the videos using a semi-manual data extractor software called Trajectory Extractor, which was developed by Lee et al. (2008). The software is highly accurate and reliable to extract trajectory data. As reported by Wong et al. (2016), the vehicle-trajectory dataset offers innovative opportunities to investigate mixed traffic characteristics, such as lane choices, lateral position distributions, spacing distributions and so on, at various levels of traffic density. Nevertheless, the method has a drawback that it is time-consuming approach and only appropriate to observe the microscopic behavior of the chaos traffic flow in a short time period.

The time step for the data extraction process was selected to capture all the changes in the drivers' behavior. Koppa (1975) described the perception and reaction time of car's drivers using a lognormal distribution with a mean of 1.3 s and a standard deviation (STD) 0.61 s. Lerner et al. (1995) recently proposed a shorter reaction time (0.54 s) for evasive maneuvering. Green (2000) reported that a fully aware driver took 0.7–0.75 s to brake. Schweitzer et al. (2007) reported experimental results for the total braking time; the mean was 0.678 s, and the STD was 0.144 s. Svetina (2016) reported that the reaction time was relatively constant across age under the single-task condition; the smallest mean was 0.59 s. For motorcycles, Tang (2003) concluded that the response duration ranges from 0.7 to 0.9 s. Minh et al. (2007) analyzed extracted data for HCM City and reported that the mean and STD of the reaction-time distribution were 0.79 and 0.24 s, respectively. Thus, the value 0.5 s is reasonable and was selected for data extraction.

2.3. Traffic flow

While collecting data, the flow was classified into two groups: motorcycles and cars. Cars in this study included all four-

wheelers, including buses and trucks. The traffic flow of each vehicle entering approach was collected, as shown in Table 2. Motorcycles dominated (~90%) for all the entry approaches. This result is similar to those of previous studies performed in Hanoi and HCM City. For example, Huynh (2016) reported a motorcycle proportion of 95.7%. Moreover, the origin-destination (OD) matrix of entry and exit at the survey site is presented in Table 3. It is easy to recognize the flow of the main roads, where traffic enters and exits along the same road. These roads contribute significantly to the total flow (>50%) and are represented by the flow from approaches 1 to 4, 2 to 5, and 3 to 6.

3. Results and discussion

In a previous study on mixed traffic at a roundabout in Taiwan, Lo (2017) only focused on macroscopic characteristics. This approach is inadequate for obtaining a comprehensive understanding of the interactions, behaviors, and mechanisms of decision-making processes. Microscopic characteristics are far more practical and meaningful for modeling vehicle movements. Therefore, the present study focused on microscopic characteristics from an individual viewpoint. This section presents the findings of the study.

Table 2 – Traffic flow entering approaches.

Approach	Duration	Traffic flow (veh/h)	Proportion (%)
1	Motorcycle	4381	91.70
	Car	348	7.29
	Others	48	1.01
2	Motorcycle	3306	87.31
	Car	441	11.66
	Others	39	1.03
3	Motorcycle	2906	91.84
	Car	222	7.02
	Others	36	1.14
4	Motorcycle	3591	90.33
	Car	342	8.61
	Others	42	1.06
5	Motorcycle	1841	90.55
	Car	165	8.12
	Others	27	1.33
6	Motorcycle	3111	90.17
	Car	309	8.96
	Others	30	0.87

Note: "Others" include buses, trucks, vans, and bicycles.

Table 3 – OD matrix of the survey site.

Entering approach	Exiting approach (%)					
	1	2	3	4	5	6
1	–	5.4	21.4	57.1	10.7	5.4
2	3.3	–	0.0	15.0	53.3	28.3
3	28.1	1.8	–	3.5	5.3	61.4
4	60.7	29.5	3.3	–	0.0	6.6
5	26.7	50.0	18.3	0.0	–	5.0
6	6.7	16.7	55.0	21.7	0.0	–

Five microscopic characteristics of motorcycles are analyzed: the acceleration and deceleration, turning angle rate and speed, critical gap, following space, and trajectory map, as shown in Fig. 5. The characteristics of the mixed traffic at the roundabout were found to be dissimilar to those of mixed traffic at mid-blocks and other types of intersections and differed significantly from those of homogeneous traffic at roundabouts.

3.1. Acceleration and deceleration

Motorcycles have a high power-over-weight ratio and are lightweight. This allows them to easily speed up or brake, which results in them continually altering their speed. The acceleration and deceleration of motorcycles were collected and plotted in a histogram, as shown in Fig. 6. The highest-frequency group was those that were slightly accelerating or decelerating (-0.4 to 0.4 m/s²); this group accounted for 74.8%. The “hard brake” and “speed up” groups accounted for small proportions, as indicated by the histogram. The remaining groups, <-1 m/s² and >1 m/s², occupied only 4.2%. These findings support the characteristics of motorcycles frequently altering their speed and emphasize that the change in speed is usually minor. The deceleration in this study ranged from 0 to -1 m/s², whereas that reported by Minh et al. (2007) ranged from 0 to -2 m/s². With regard to acceleration, the results of the two studies were similar: they mostly ranged from 0 to 1 m/s². This could be because the distinctive speed at roundabouts, which is 3.61 m/s on average (almost half of the average speed at mid-blocks, i.e., 6 m/s). Lower speeds result in a lower required deceleration rate under the same period of time to stop. Additionally, maneuverability is higher at lower speeds, allowing looser time to react. Therefore, drivers prefer soft braking at roundabouts to hard braking.

The acceleration-deceleration and speed profiles of a single motorcycle are shown in Fig. 7. In the first 20 s, the speed continuously fluctuated between 0 and 0.4 m/s. This indicates that the driver faced complex situations and was continuously accelerating or decelerating to respond. Consequently, the motorcycle exhibited continuous changes in both its moving direction and its speed.

The positions at which motorcycles often speed up or brake were considered. Thus, the acceleration and deceleration scatters were plotted along the trajectory. An example of motorcycles traveling from approach 1 to approach 4 is shown in Fig. 8. The color in the diagram represents the magnitude of the speed change. The results indicate that at the circular flow, they motorcycles did not exhibit hard braking or quick acceleration; rather, they decelerated and slightly accelerated. At this location, the motorcyclists had already matched their speed with the flow; therefore, there were few significant speed changes.

3.2. Turning angle rate and speed

The turning angle represents the change in the moving direction within the local frame of reference. It is not the steering angle of the handle or the front wheels but the angle between the current and previous moving directions. Ma et al. (2017) stated that it is one of the key parameters of the turning process. The turning angle describes the magnitude of the change in the moving direction. It was described by Lee (2007) and was called the “veering angle” by Lenorzer et al. (2015) and the “angular velocity” by Ma et al. (2017). However, in none of these studies was this quantity calibrated or modeled. The present paper presents the first derivative of the turning angle, which is called the turning-angle rate (T.A.R). The T.A.R is the rate at which a vehicle changes its moving approach. It is calculated as follows

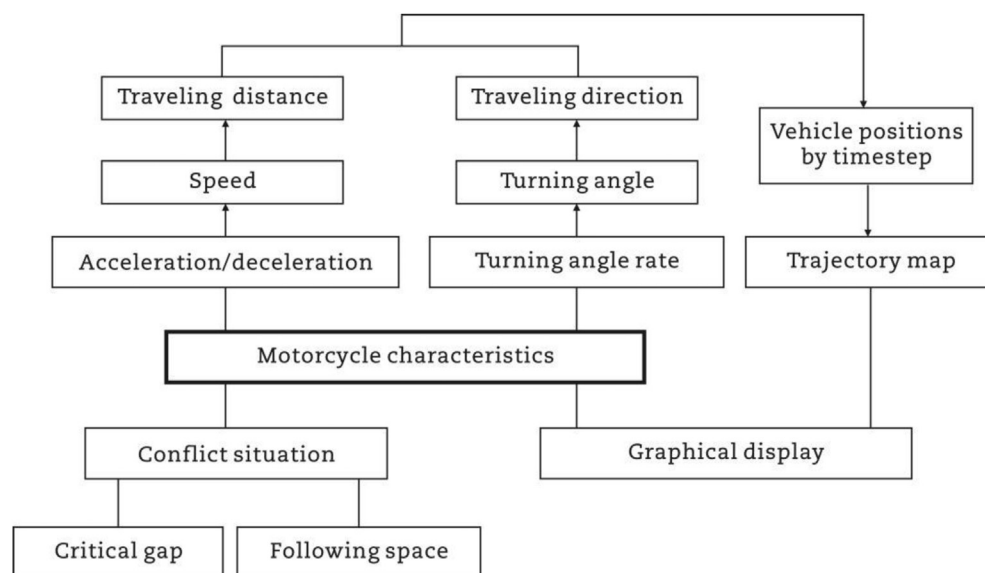


Fig. 5 – Conceptual framework of the analyzed microscopic characteristics.

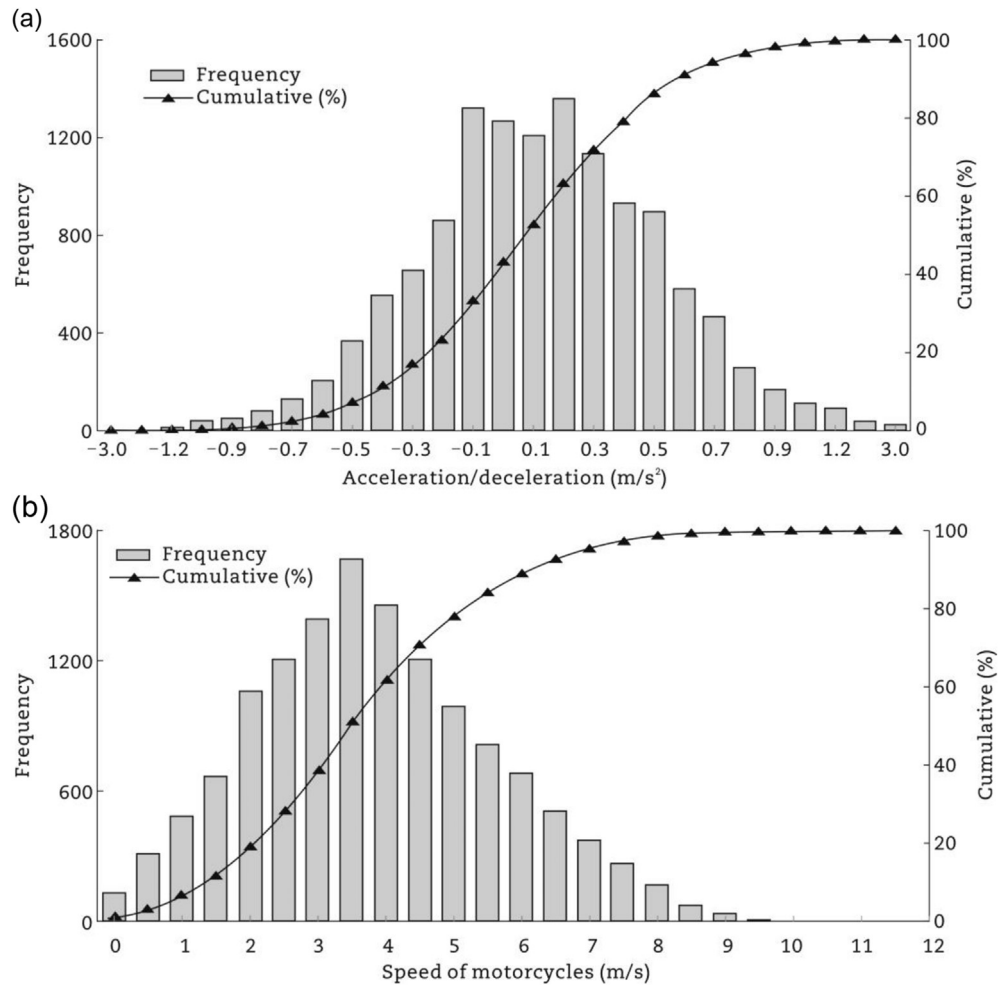


Fig. 6 – Histogram of the acceleration/deceleration, and speed of motorcycles. (a) Acceleration/deceleration. (b) Speed of motorcycles.

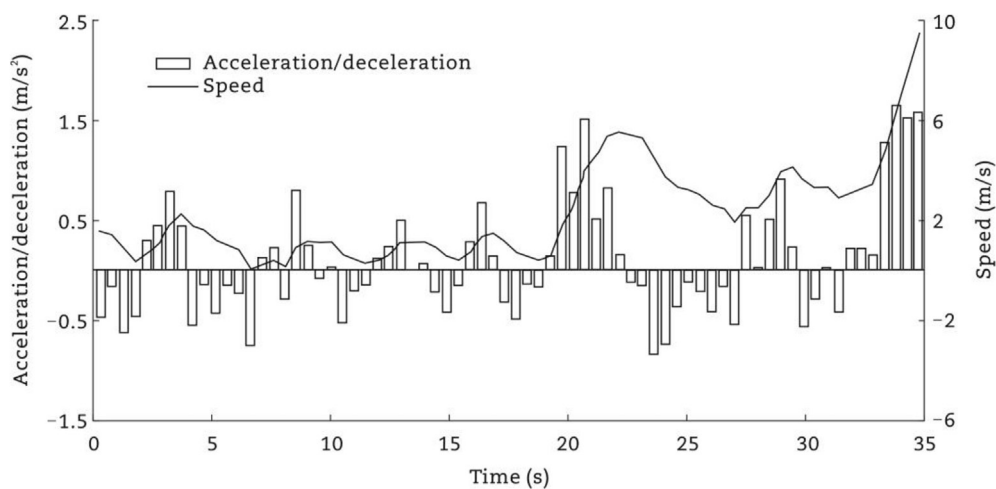


Fig. 7 – Acceleration/deceleration and speed profiles of a motorcycle.

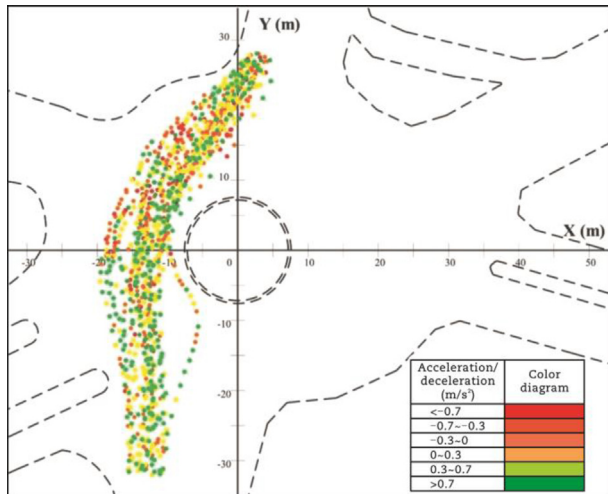


Fig. 8 – Acceleration and deceleration scatter plots.

$$\text{T.A.R} = |\text{dir}_{\text{current}} - \text{dir}_{\text{previous}}| / \text{time_step} \quad (1)$$

The unit for the T.A.R is degrees per second (degree/s). The $\text{dir}_{\text{current}}$ represents the current moving direction in the global coordinate system. It is the angle between the unit vector which point of the north of the map and the speed vector of the vehicle at a given time, t . Its value is in the range $[0, 360]$. The $\text{dir}_{\text{previous}}$ represents the previous moving direction in the global coordinate system. It is the angle between the unit vector point of the north of the map and the speed vector of the vehicle at the previous calculated frame, i.e., at time t (time_step). The time_step (s) represents the period of time between two frames that are used for tracking the vehicle's position.

This paper presents the first examination of the relationship between the T.A.R and the speed. The maximum value of the T.A.R, which was dependent on the traveling speed, was formulated. This inspection can facilitate the calibration in future modeling studies. Fig. 9 shows more than 14,000

coupled data points for the T.A.R and speed. Generally, at low speeds, the T.A.R is high; the inverse is also true. At high speeds, the maneuverability of motorcycle is limited by physic rules; e.g., the inertial restriction. The driver intends to perform large-angle maneuvers at a low speed. This is also shown in Fig. 9. Thus, the boundary of the data, rather than the central tendency, is the main focus of this investigation.

To clarify the relationship between the T.A.R and the speed, the data were first filtered by eliminating the most dispersed part (~5%). As the boundary of the T.A.R was the main focus, 95% of the data were considered to be valid. Binning the speed into small intervals (of 0.5 m/s) allowed the data distribution to be calculated more easily. In each interval, the representative horizontal value was the mean speed of the segment. In the vertical direction, the values of 95% of the cumulative distribution of the turning angle rate were considered to be representative values; they are displayed as red circles in Fig. 9. These values were employed to estimate the fitting curve. The results of regression analysis and curve estimation, which were performed using the IBM Statistical Package for the Social Sciences (SPSS) software, are presented in Tables 4 and 5.

Owing to the roughness of the data, the power curve and the exponential curve were estimated. The T.A.R and speed exhibited a very strong correlation in both cases, with R values of 0.929 and 0.776, respectively. The R -squared value for the power model was significantly higher than that for the exponential model (0.864 vs. 0.603). Thus, the power model provided a more accurate estimation. The high adjusted R -squared value (0.856) confirmed that the power curve fit well

Table 4 – SPSS regression analysis results.

Curve estimation	R	R-squared	Adjusted R-squared	Std. error of estimate
Power	0.929	0.864	0.856	0.188
Exponential	0.776	0.603	0.581	0.321

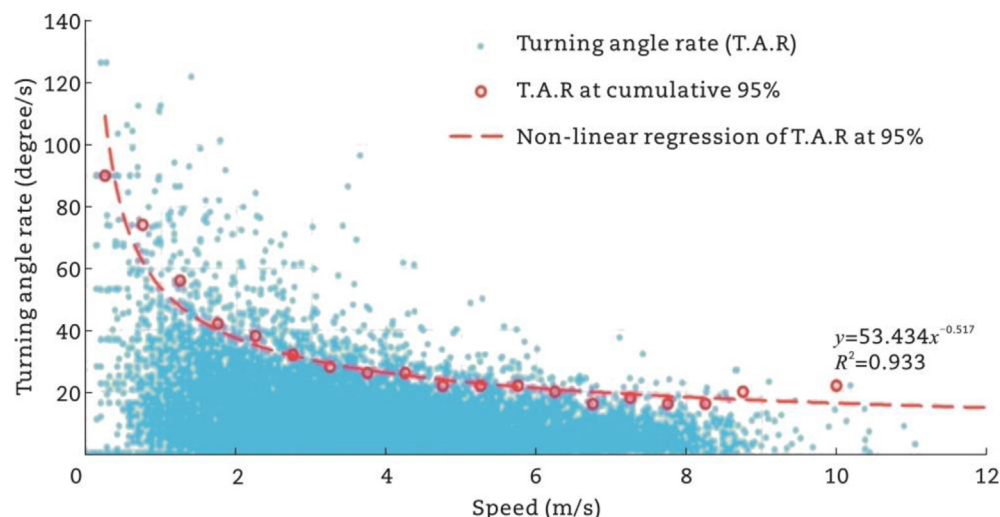


Fig. 9 – Relationship between the T.A.R and speed.

Table 5 – Coefficients of power regression curve estimation using SPSS.

	Unstandardized coefficients		Standardized coefficients	t	Sig.
	B	Std. error	Beta		
Independent variable (speed)	–0.125	0.032	–0.776	–5.227	0.000
Constant	51.895	7.301		7.108	0.000

Note: the dependent variable is T.A.R.

with the data. The standard error of the estimation was minimal compared with the range of the T.A.R. (0.188 vs. 140). This suggests that the error for the power-curve estimation was small. Moreover, the significance levels of the variables were close to zero and were significantly smaller than 0.05. Thus, all of our B coefficients were statistically significant. The estimated curve (the red dashed line) and its formula are shown in Fig. 9. The maximum value of the T.A.R. can be calculated as follows

$$\text{T.A.R.} = 51.895v^{-0.125} \quad (2)$$

where v represents the current speed of the vehicle (m/s).

The size of the roundabout also influenced the T.A.R.; a smaller central island resulted in a more significant effect of the circulating movement on the vehicle turning angle. Thus, the T.A.R. of the same vehicle is higher in a smaller roundabout. For determining the maximum T.A.R. (which was dependent on the speed) and utilizing this effect, the selected site, i.e., a small-sized roundabout, was appropriate.

To validate the estimated curve, the data-splitting technique Snee (1977), which is one of the accredited methods for obtaining a good internal validation Giancrisotofaro and Salmasso (2003), was employed to divide the samples into two sets. Fitting samples accounted for 75% of the original samples, and the other 25% of the samples were validation samples. Four goodness-of-fit measures, which were used for measuring simulation of heterogeneous traffic, were also utilized: the mean absolute percentage error (MAPE) (Mullakkal et al., 2015; Swamidass, 2000), mean absolute deviation (MAD) (Swamidass, 2000), root-mean-square error (RMSE) (Toledo and Koutsopoulos, 2004; Dowling et al., 2004) and root mean square normalized error (RMSNE) (Hourdakis et al., 2003; Toledo et al., 2003; Ma and Abdulhai, 2002). Especially, the MAPE and MAD were used appropriately to evaluate the goodness-of-fit between the predicted and actual data by Swamidass (2000).

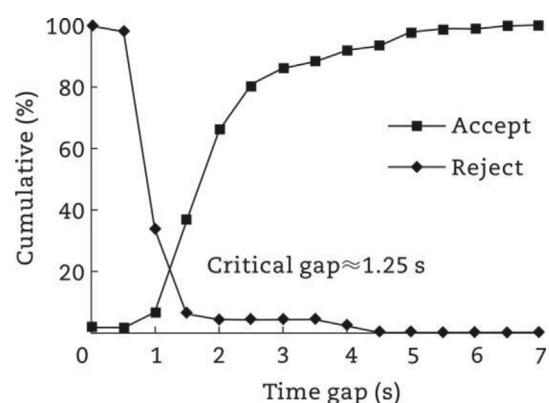
The validation results for the estimated curve are presented in Table 6. According to Lewis' scale of interpretation for the MAPE, the model was validate as good, as the MAPE (18.8%) was in the range of 11%–20% (Lewis, 1982). The RMSE and RMSNE, which were 1.57 and 0.45, respectively, indicated that the overall performance of the estimated curve was good. The MAD (4.311) was small compared with

the range of the data (from 0 to >100), confirming the accuracy of the estimated curve.

3.3. Critical gap

The critical gap and follow-up headway are the major gap–acceptance parameters that explain the interaction of the merging flow and circulating flow at roundabouts Giuffrè et al. (2016). A gap is an opening distance or time in the circulating flow that an entering vehicle can exploit. When the driver of the entering vehicle refuses to merge with the circulating flow, the gap is denoted as a “rejected gap.” When the driver considers the gap safe enough to merge, the gap is denoted as an “accepted gap” Shaaban and Hamad (2018). The critical gap refers to the specific gap size for which a driver would accept if it were larger and would reject if it were smaller. Even though the accepted gap has been used as the critical gap for estimating the capacity at roundabouts (TRB, 2010), it is unreasonable to ignore the rejected gap (Fitzpatrick et al., 2013). Thus, in the present study, Raff's method (Raff and Hart, 1950) was used to estimate the median critical gap, in accordance with the spatial and temporal gap analysis approach of Fitzpatrick et al. (2013). Owing to the nature of non-lane-based vehicles, the follow-up headway was unsuitable and was replaced with the following space defined in Section Following space.

The critical gap of a motorcycle facing a motorcycle is shown in Fig. 10. The cumulative line of the accepted gap rises sharply in the range of 1–2.5 s. The majority of drivers (80.23%) accept the gap when it is >2.5 s. The cumulative line of the rejected gap falls significantly, even in the shorter range of 0.5–1.5 s. Most of the drivers (93.18%) reject the gap when it is <1.5 s. These shapes are similar to those observed by Dodappaneni et al. (2016). It can be inferred that motorcycles are highly

**Fig. 10 – Critical gap for a motorcycle facing a motorcycle.****Table 6 – Validation results for the estimated curve.**

Estimated curve	MAPE	MAD	RMSE	RMNSE
Power	0.188	4.311	5.089	0.031

sensitive to small gaps. The critical gap is the crossing point of the cumulative lines of the accepted and rejected gaps. The critical gap in the present study (1.25 s) was shorter than those previously reported [Dodappaneni et al. \(2016\)](#); [Arroju et al. \(2015\)](#). The reason could come from the conclusion of former studies ([Dodappaneni et al., 2016](#); [Patel and Khode, 2016](#)) that the lower critical gap was found under smaller-traffic volume conditions. The motorcycle volume in this research is about triple compared to [Dodappaneni et al. \(2016\)](#). In addition to the traffic volume, the traffic composition affects the critical gap. With a higher proportion of motorcycles, the critical gap is smaller. The proportion of motorcycle in this study is higher from 4.5%–36.3% compared to [Dodappaneni et al. \(2016\)](#). This once again emphasizes the advantages of motorcycles with regard to utilizing a small gap. Thus, it is concluded that a large traffic volume with the dominance of motorcycles results in a small time gap.

3.4. Following space

The “follow-up time,” which is also called the “follow-up headway” in spatial measurements [Zheng et al. \(2012\)](#), is the longitudinal time gap between two queued vehicles entering a roundabout [Dahl and Lee \(2012\)](#). When considering the following behavior of non-lane-based vehicles, the longitudinal gap is insufficient. Thus, in the present study, both the longitudinal and lateral gaps related to the following space are considered.

Motorcycles have unique following behavior: they usually maintain a clear space around themselves. This behavior was

first described in a study on mixed traffic characteristics in mid-blocks [Lee et al. \(2009\)](#). That study emphasized that motorcycles follow cars with an extremely small safety headway when traveling close to the edge of the car and with a larger headway when aligned with the center. Thus, the minimum following space from the rear of the preceding vehicle has the shape of a triangle. Even though this following space is defined by the preceding vehicle, the paper pointed out the distinctive shape of the following space. [Nguyen et al. \(2012\)](#) replaced the safe headway with a safety space, which had a half-ellipsoid boundary. The size of the ellipse depends on the traveling speed, the reaction time, the vehicle's dimensions, the types of vehicles, the longitudinal and lateral distances to the influential vehicle [Nguyen et al. \(2012\)](#).

This raises the question of how the following space changes when motorcycles approach roundabouts, rather than mid-blocks. We attempted to answer the foregoing question by calibrating the following space at the studied site. Motorcycles were observed when they performed the following behavior. Other behaviors (e.g., when they faced conflicts) were not considered. The relative positions of the surrounding vehicles were accumulated via time series, as shown in [Fig. 11](#). The data were categorized into three groups according to speed. The first, second, and third groups' initial speeds were 0–2.78 m/s (0–10 km/h), 2.78–4.17 m/s (10–15 km/h), and >4.17 m/s (>15 km/h), respectively. In each time, only motorcycles at the first boundary, which had a significant influence on the subject vehicle, were tracked. The others, which were hidden from view, were neglected.

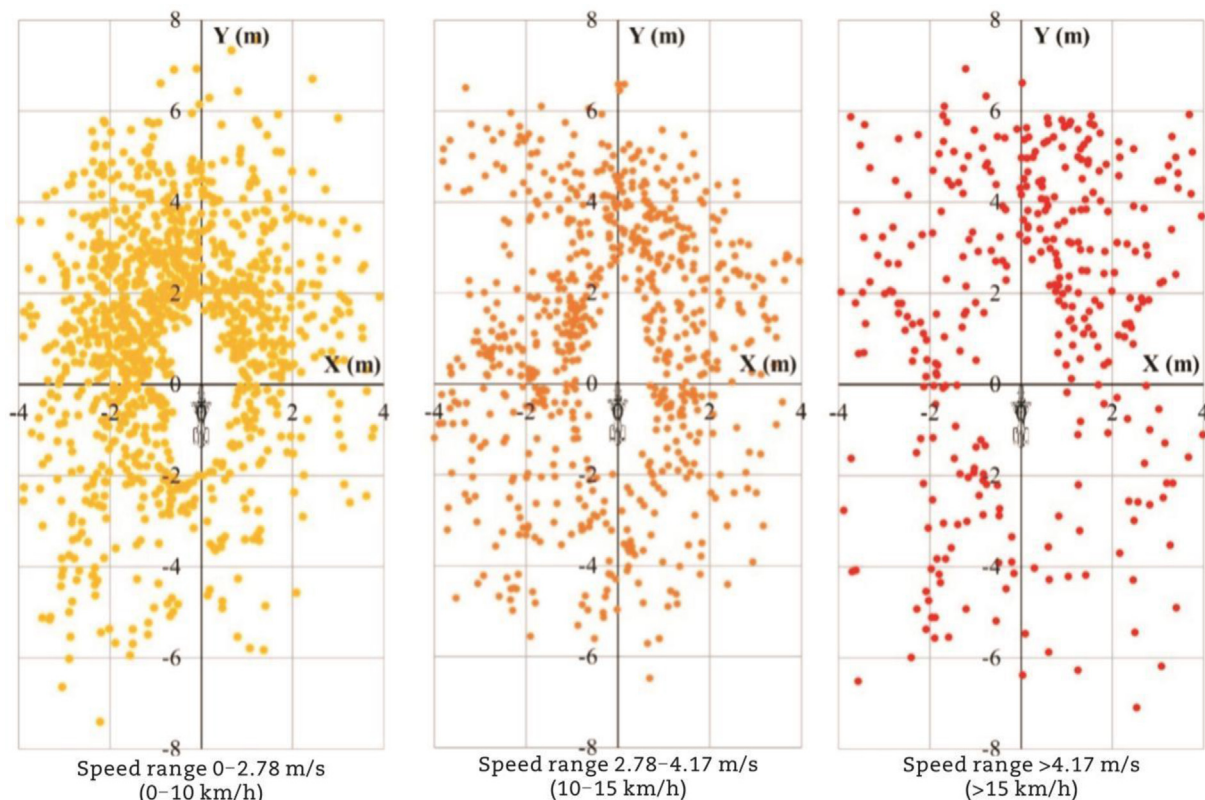


Fig. 11 – Following space of the motorcycle.

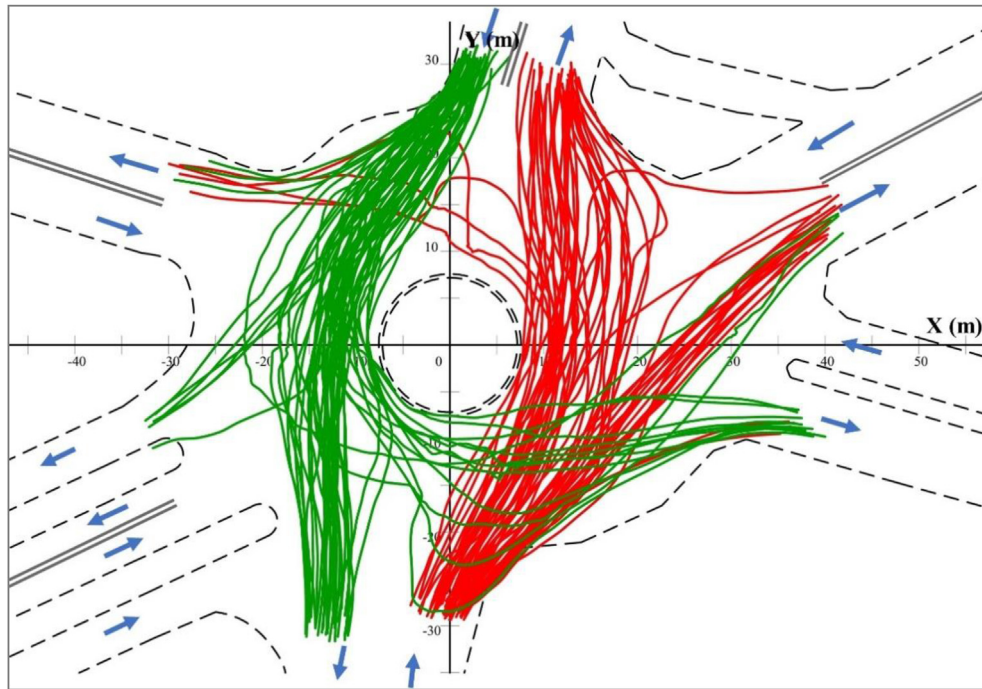


Fig. 12 – Trajectory maps of approaches 1 and 4.

The distributions of the relative positions are shown in Fig. 11 in the local Cartesian coordinate systems of the subject motorcycle. All the markers in each of the graphs were tracked from the middle-front point of the vehicle. The middle-front point of the subject motorcycle is represented by the origin (0, 0). The oval-shaped clearance in the middle of three of the graphs supports the theory of safety space. The size of this space increases as the speed increases. These results agree with the dynamic lane width reported by Minh et al. (2007) across these three speed ranges. However, compared with the size of the safety space suggested by Nguyen et al. (2012), the lateral dimension was larger and was independent of the speed (1.5 m compared with 2.6 m). The longitudinal dimension was larger at higher speeds and was also larger than that of the safety space (1.74 and 2.88 m at speed range 0–2.78 m/s and 2.78–4.17 m/s, compared with 1.39 and 2.08 m at the same speed range for the safety space).

This distinction is attributed to the speed and road type. The average speed in this study was 3.61 m/s, whereas that calibrated by Nguyen et al. (2012) was 8.33 m/s. Even though both studies were conducted in HCM City, the survey site in the present study was a roundabout, whereas previous studies focused on mid-blocks. Moreover, in the low speed range, vehicles were denser near the boundary of the clearance, compared with the higher speed range. At higher speeds, the drivers preferred to maintain the desired gap, rather than the minimal gap. Therefore, the density at the following space's boundary was lower than that at lower speeds.

3.5. Trajectory map

A trajectory map was drawn by connecting the consecutive positions of the vehicles over time. Most of the vehicle trajectories were smooth, indicating that they were reasonable.

Fig. 12 shows the trajectory maps of 120 motorcycles from approach 1 (green path) and approach 4 (red path). The trajectories were dispersed and covered the entire width of each approach when entering or exiting. The lateral position of each motorcycle at any cross section could be easily inferred. Moreover, the trajectories were complex and unpredictable. This is in accordance with the unique characteristics of motorcycles, which frequently maneuver inside the roundabout; however, it presents a challenge for conventional models to reproduce these trajectories.

4. Conclusions

Mixed traffic is attracting increasing research attention owing to its prevalence in developing countries. In this study, academic contributions were made to the comprehension of mixed traffic at roundabouts. The traffic state in Vietnam (motorcycle-oriented mixed traffic) and the microscopic characteristics at roundabouts were analyzed. Associated with the first contribution, the two fundamental factors that characterize mixed traffic are the performance rule and the appearance of small vehicles. Traffic that lacks one of these factors cannot be identified as mixed traffic. Vietnam was selected as a case study, as it exhibits both factors. Even though the country's traffic is distinctive in that it is dominated by motorcycles (90%), it is representative of mixed traffic. The comprehension of the mixed traffic in Vietnam presented herein supports the understanding of the traffic states in other mixed traffic countries.

Regarding the second contribution, the microscopic characteristics of motorcycles were emphasized. The acceleration, deceleration, turning angle rate, critical gap, following space, and trajectory map were analyzed in detail. According to the

results, three special microscopic characteristics of motorcycles were identified. First, motorcycles continuously alter their speed and direction, usually during travel. This study built on previous studies by incorporating a quantitative analysis. Second, the maneuverability of motorcycles is limited by their speed. The relationship between the turning angle rate and the speed was formulated in an equation, allowing for its direct application to micro-simulations. Third, the critical gap of motorcycles was small in the study case: 1.25 s. Fourth, motorcycles maintain a clear space around them. This finding supports those of previous studies on mid-blocks and enhances our understanding of the psychology of motorcycle drivers at roundabouts. The results of this study can be employed for the development of models and for parameter estimations using simulation software.

In spite of these contributions, the study had several limitations. First, it was restricted to one study site. In future studies, the variety of geometrical designs should be expanded, for example, the inscribed diameter, central island, and number of legs. Additionally, the characteristics of other vehicle types should be considered. Moreover, the safety space herein focused only on same-flow vehicles. The effects of conflicting space from crossing flows on the subject motorcycle should be estimated. Finally, the characteristics of cars and other lane-based vehicles were not considered. The interactions between motorcycles and other lane-based vehicles should be considered in future studies.

Conflict of interest

The authors do not have any conflicts of interest with other entities or researchers.

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gent transportation systems.

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